MIL-T-41812H
30 June 1988
SUPERSEDING
MIL-T-41812G
22 June 1984

MILITARY SPECIFICATION

TENT LINER SECTIONS, FRAME-TYPE MAINTENANCE, MEDIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers flame and mildew-resistant cotton sateen tent liner sections with covers. The liner consists of end sections and insulated intermediate sections. Each section is provided with a cover for protection during transport.

1.2 Classification. Tent liner sections shall be of the following types as specified (see 6.2).

Type I - End section "A", 4 feet wide
Type II - End section "B", 4 feet wide
Type III - Intermediate section, 8 feet wide

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
T-T-871	-	Twine, Cotton, Wrapping
T-T-911	-	Twine, Fibrous, Jute
V-F-106	-	Fasteners, Slide, Interlocking
V-T-285	-	Thread, Polyester
QQ-A-601	-	Aluminum Alloy Sand Castings
QQ-B-613	-	Brass, Leaded and Nonleaded: Flat Products (Plate Bar, Sheet and Strip)
RR-C-271	-	Chains and Attachments, Welded and Weldless
CCC-C-429	-	Cloth, Osnaburg, Cotton
CCC-C-467	-	Cloth, Burlap, Jute (or Kenaf)
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood

MILITARY

MIL-L-1709	-	Lines, Tent
MIL-C-2399	-	Cement, Liquid, Tent Patching
MIL-R-3390	-	Rings, Dee
MIL-G-16491	-	Grommet, Metallic
MIL-R-30500	-	Rope, Polyester
MIL-C-43122	-	Cloth, Sateen, Cotton, Flame Resistant Treated
MIL-C-43256	-	Cord, Polyester, Solid Braid
MIL-T-43566	-	Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors
MIL-C-43627	-	Cloth, Duck, Cotton, Plied Yarns, Fire, Water, Weather and Mildew Resistant Treated, Lightdry Finish
MIL-W-44049	-	Webbing, Textile and Tape, Textile, Polypropylene, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

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* (Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

Department of the Army Field Manual FM 10-16
General Repair of Tents, Canvas and Webbing

* (Copies may be obtained from the Superintendent of Documents, U.S. Government Printing office, Washington, DC 20402.)

DRAWINGS

* US ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

	5-4-686	- Tent Liner Sections, Frame-Type, Maintenance, Medium; End Section, Miscellaneous Details
	5-4-688	- Intermediate Section Assembly Details and Sections
	5-4-689	- End Section "A" and "B"
	5-4-691	- Erection Instructions
	5-4-703	- Cover for Intermediate Section
*	5-4-3359	- Tent Cover

* (Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 241	- Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube
D 3951	- Standard Practice for Commercial Packaging

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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* (Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.5).

3.2 Materials and components. Materials and components shall be as specified herein (see 6.7).

* 3.2.1 Cloth, sateen, cotton. The cloth for the tent liner sections shall conform to class 1 of MIL-C-43122. Requirements for shrinkage shall not apply.

3.2.2 Cloth, duck, cotton. The cloth for the covers shall conform to MIL-C-43627, Olive Drab 7.

3.2.3 Tape, cotton. The tapes shall conform to type I, class 1a, 3/4 inch, 1 inch, and 2 inch widths of MIL-T-43566 and the shade shall be natural. Alternatively, the tape shall be polypropylene conforming to type I, class 3 of MIL-W-44049.

* 3.2.4 Thread, polyester.

* 3.2.4.1 Liner. The thread for the liner shall conform to type I or II, class 1, sub-class A, color natural of V-T-285.

* 3.2.4.2 Cover. Thread for the cover shall conform to type I, class 1, sub-class B of V-T-285 and the color shall be Olive Drab S-1 (C.A. 66022).

* 3.2.4.3 Sizes. The thread sizes shall be as follows:

For all stitching on tent liner, except stitching indicated by "Z" on drawings	- Size E for needle and bobbin
For stitching on tent liner indicated by "Z" on drawing	- Size E for needle, size B for looper
For all stitching on cover except stitching indicated by "Z" on drawings	- Size F for needle and bobbin
For all stitching on cover indicated by "Z" on drawings	- Size F for needle, size E for looper
For bartacking and zig-zag stitching	- Size E

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3.2.5 Cord, polyester. The polyester cord shall be natural in color, 1/8 inch diameter and shall conform to MIL-C-43256.

3.2.6 Rope, polyester. The rope shall conform to 3/4 inch circumference (1/4 inch diameter) and 5/8 inch circumference (3/16 inch diameter) of MIL-R-30500.

3.2.7 Plastic film. The unsupported polyethylene film used for the window panes shall conform to the requirements for 0.0060 inch thick film, type I, class 1, grade A, B, or C, finish 1 of L-P-378 except that the thickness shall be 0.020 ± 0.004 inch. Requirements for watervapor permeability, slip, ink adhesion, impact resistance and heat-seal strength are not applicable.

3.2.8 Lines, tent. The cover tie lines shall conform to type XXI of MIL-L-1709.

* 3.2.9 Insulation. The insulation material shall be composed of glass fibers firmly bonded with a thermosetting plastic resin into a resilient mat structure 1-1/4 inches thick + 3/4 inch, -0 inch tolerance having a weight of 2.1 pounds ± 10 percent per 24 square feet when tested as specified in 4.4.1.1. The batting bulk density range shall be 0.47 to 0.92 pounds per cubic foot.

3.2.9.1 Compressional recovery. The compressional recovery of the insulation material shall be not less than 90 percent when tested as specified in 4.4.1.1.

3.2.10 Grommets, metallic. The grommets shall conform to type III, class 3, size Nos. 1,2,3, and 4 of MIL-G-16491, and shall be located where indicated on drawings.

* 3.2.11 Fastener, slide, interlocking. The slide fasteners shall conform to type I, style 2, size H, wire stirrup pulls of V-F-106. Slide fastener chain shall be fabricated of brass or polyester coil. The slider for the brass chain shall be brass and the slider for the polyester coil shall be die cast zinc alloy. The slide fastener tape shall be dyed to match Olive Drab shades S (CA 66519) and be water repellent and mildew-resistant. (Tapes made of 100 percent polyester need not be tested for mildew resistance.) The color of the tape after treatment will be considered acceptable. The tape shall show good fastness to weathering. Alternatively, the slide fastener chain shall be polyester continuous monofilament in a ladder or coil type configuration. Color of the chain shall be olive drab or natural finish brass and the sliders shall be brass.

3.2.12 Rings, dee. The dee rings shall conform to class 1 or 2, configuration A of MIL-R-3390.

3.2.13 Hook, "S". The "S" hook shall be galvanized, steel wire, and shall conform to the design and dimensional requirements specified on Drawing 5-4-686 (see 6.3).

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3.2.14 Chain, machine, twist link. The chain shall conform to type I, grade C, class 5, style 2, 1/8 inch nominal size, hot dipped galvanized of RR-C-271.

3.2.15 End, eye. The eye ends shall be malleable iron, galvanized, 3/8 inch and 1/2 inch sizes (see 6.4). As an alternate, the end eye shall be made of aluminum alloy conforming to 356.0 T6 of QQ-A-601.

3.2.16 Clamp, rope, brass. The clamps which are used to secure the ends of polyester rope (alternate method) shall conform to copper alloy No. 260, sheet or strip of QQ-B-613 in the design and dimensions shown on Drawing 5-4-703 (see 6.4).

3.2.17 Pipe, aluminum-alloy, seamless. The pipe shall be 6061-T6, 3/8 inch and 1/2 inch nominal pipe size, standard (schedule 40), conforming to ASTM B 241.

3.2.18 Twine, cotton. The twine shall conform to type II, class 1, 5 ply of T-T-871.

3.3 Construction. The construction shall conform in all respects to the drawings listed in section 2 and as specified herein.

3.3.1 Stitching, machine.

* 3.3.1.1 Types of stitching. All stitch types except bartacking shall conform to FED-STD-751 as follows:

For all stitching on liner sections and covers, except zig-zag stitching, stitching indicated by "Z" on drawings, and alternatively for attachment of slide fasteners	- Type 301, 5 to 7 stitches per inch
For all zig-zag stitching on liner sections	- Type 304, 1/8 to 3/16 wide (bight), 10 to 14 stitches per inch
For stitching on liner sections and covers indicated by "Z" on drawings and alternatively for attachment of slide fasteners	- Type 401, 5 to 7 stitches per inch Chain portion of stitching shall not appear on outside of liner sections and cover.
For bartacking	- 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch wide (bight) 28 stitches per inch

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* 3.3.1.2 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/2 inch minimum except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in puckering of material sewn. The lock shall be embedded in the material sewn.

* 3.3.1.2.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch behind the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch before the defective area (1/2 inch on box, box and cross and W-W stitching) and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

* 3.3.1.3 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching. Both ends of seams or stitching produced with a 401 stitch type, when not turned under a hem or held down by other stitching, shall have a 1/2 to 3/4 inch chain extending beyond each end. The looper (under thread) shall always be on the inside of the tent or cover. All repairs shall be made using a 301 stitch in accordance with 3.3.1.2.1a or 3.3.1.2.1b.

* 3.3.1.4 Type 304 stitching. For type 304 stitching, thread tension shall be maintained so that there will be no loose or excessively tight stitching. Repairs shall be in accordance with 3.3.1.2.1a or 3.3.1.2.1b except that the stitching shall over stitch the defective stitching area a minimum of three stitches onto the existing stitching. At least 1/4 inch of backstitching, overstitching or an increased number of stitches per inch shall be used to secure the ends of stitching.

* 3.3.1.5 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements of the stitch pattern, stitches per inch, size and type of thread are met, and at least three or more tying, overlapping or backstitches are used to secure the ends of the stitching.

* 3.3.1.6 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a length of not more than 1/4 inch.

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* 3.3.1.7 Lubrication of thread. The addition of any lubricant to the polyester sewing thread prior to or during the sewing operation is prohibited.

* 3.3.1.8 Stitching margins. Unless otherwise specified, all stitch margins shall be 1/8, +1/16 -0 inch.

* 3.3.1.9 Bartacking. Bartacking shall be free of thread breaks or loose stitching.

3.3.1.10 Becket line. The 304 stitching attaching the becket line to the roof portion shall pass through the cord.

* 3.3.2 Splicing of beckets. Splicing will be permitted in the liner sections and if performed, shall conform to Drawing 5-4-688 with a minimum distance of 10 feet between splices. No splice shall be in any portion of the loop.

3.3.3 Attachment of slide fasteners. Care shall be exercised to trim all thread ends and maintain specified distances between edges of fabric attached to slide fastener tape so that there will be no interference with the operation of the slider.

* 3.3.4 Setting of grommets. Holes punched to receive the grommets shall be smaller than the grommet barrel so that the barrel must be forced through the hole. Care shall be taken to see that washers are placed on the outside face of end and intermediate sections. The grommets shall be securely clinched without cutting the materials. Roll of the clinched grommet shall not contain splits.

3.3.5 Insulation. The insulation shall be one continuous piece for each intermediate and end section; i.e., piecing is not permitted.

3.3.6 Tufting. Tufting shall be performed as specified on the drawings using two strands of 5 ply twine, specified in 3.2.18, twisted together.

* 3.3.7 Splicing of cloth panels. One splice per panel shall be allowed. Splices in adjoining panels shall not be placed adjacent to each other. No piece shall be less than 1 yard long. The method of splicing shall be as shown on Drawing 5-4-688.

3.3.8 Repairs. Any necessary repairs shall be effected in accordance with Department of the Army Field Manual FM 10-16, except that hand-sewn patches, or cemented patches other than as specified herein, will not be acceptable. Cemented patches on holes, cuts or tears not more than 1 inch in diameter or length, using cement specified in MIL-C-2399, shall be permitted. A maximum of two cemented patches per tent liner section shall be allowed without prior authorization. All other repairs shall require contracting officer authorization.

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* 3.3.9 Replacement of defective components. During the spreading, cutting and manufacturing process, components of the liner having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.

* 3.3.10 Fusing of ends of synthetic cord, webbing and tape. All ends of synthetic cord, webbing, and tape shall be fused. The apparatus used to fuse the ends shall be capable of providing sufficient heat to provide a smooth edge and with the cut ends of the yarns all fused together. Fusing of the ends shall be accomplished prior to being assembled for stitching.

3.4 Marking. All marking and labels shall conform to DDD-L-20.

3.4.1 Identification label. Identification labels shall conform to type VI, class 6, and shall be 3-1/2 inches by 5-1/2 inches minimum, and a blank space of at least 1-1/4 inches shall be provided on the bottom of the label for the Department of Defense inspector's stamp. Identification labels shall be stitched to the end sections, intermediate sections and covers in the locations shown on the drawings.

* 3.4.2 Erection Instruction label. Erection instruction labels shall be type VI, class 7 and shall be stitched to the end sections, intermediate sections, and covers in the locations shown on the drawings. The label shall be a proportionate copy of Drawing 5-4-691, as reproduced from a glossy print of the drawing furnished by the contracting officer. The size of the label shall be, borderline to borderline, 15 $\pm$ 1/4 inches in the longer dimension with a minimum 1/2 inch margin outside of the borderline on all sides.

3.4.3 Special marking. The letters "US" and the nomenclature on the end sections, intermediate sections, and covers shall be type IV, class 9 in the size characters and locations shown on the drawings. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

3.4.4 Warning instruction label. Warning instruction labels shall be type VI, class 9 and shall be stitched to the end sections and intermediate sections in the locations shown on the drawings. The warning label shall read as follows:

WARNING
WHEN TENT IS CLOSED, EXHAUST OF INTERNAL COMBUSTION ENGINES
MUST BE VENTED TO THE OUTSIDE

The word "WARNING" shall be in red letters at least 2 inches high and the remainder of the label shall be black letters at least 1/2 inch high. Fastness of the class 9 labels shall be as specified for class 7 labels.

* 3.5 Workmanship. Care shall be taken in fabricating blankets to see that insulating material lies flat within the cover, is not buckled or doubled up, and is laid adjacent to seams as illustrated on applicable drawings. Care shall be taken in sewing so that no needle chews occur. Thread ends shall be trimmed

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to 1/2 inch or less in length. Metal components shall be free of burrs, sharp edges, or corroded areas and not be broken or malformed. The liner shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4, and tested for the characteristics specified in 4.4.1.1. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Insulation testing. The insulation shall be tested for the characteristics listed in table I. The lot size shall be expressed in units of 1 pound. The sample unit shall be 24 square feet of insulation. The inspection level for each characteristic listed shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5. All requirements are applicable to the sample unit.

TABLE I. Insulation Tests

Characteristics	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Weight	3.2.9	Direct weighing	1	Pass or fail
* Thickness	3.2.9	4.5.1	3	Nearest 1/16 inch
Compressional recovery	3.2.9.1	4.5.2	5	Nearest 1 percent

* 4.4.2 In-process inspection. Inspection shall be made of the following operations or assemblies to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation. Items which cannot be corrected shall be removed from production.

a. Holes punched to receive grommets are in compliance with 3.3.4.

b. There is no lubrication of sewing thread prior to or during sewing operations in compliance with 3.3.1.7.

c. Insulation material for intermediate and end sections is one continuous piece in compliance with 3.3.5.

d. Two strands of 5-ply twine are used in tufting operations and the twine strands are thoroughly twisted in compliance with 3.3.6.

* 4.4.3 End item visual examination.

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* 4.4.3.1 End section ("A" or "B"). The end item shall be in a position so that both the inside and outside of the end section can be examined for the defects listed in tables II and III. The lot size shall be expressed in units of end sections. The sample unit shall be one end section. The inspection level shall be I, and the AQL, expressed in terms of defects per hundred units, shall be 10 for major defects, 25 for major and minor A combined defects, and 100 for total (major and minor combined) defects.

* 4.4.3.2 Intermediate section. The end item shall be in a position so that both the inside and outside of the intermediate section can be examined for the defects listed in tables II and IV. The lot size shall be expressed in units of intermediate sections. The sample unit shall be one intermediate section. The inspection level shall be I, and the AQL, expressed in terms of defects per hundred units, shall be 6.5 for major defects, 15 for major and minor A combined defects, and 65 for total (major and minor combined) defects.

* 4.4.3.3 Covers only. The end item shall be in a position so that both the inside and outside of the cover can be examined for the applicable defects listed in table II. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be I, and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects, 10 for major and minor A combined defects, and 40 for total (major and minor combined) defects.

TABLE II. End item visual defects.

Examine	Defect	Classification Major	Minor A	Minor B
Fabric	Cut, tear, or hole:			
	- greater than 1/4 inch in length or diameter	X		
	- greater then 1/8 inch but not greater than 1/4 inch in length or diameter		X	
	- more than five holes 1/8 inch or less in length or diameter			X
	Broken or missing yarn, or multiple floats clearly visible.		X	
Tie tapes	Edge frayed or scalloped		X	
	Stitched ends of tie tapes not turned under as specified		X	
Facing tapes	More splices per component than allowed			X
Hardware	Broken	X		
	Malformed:			
	- fails to perform intended function	X		
	- but will perform intended function		X	

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TABLE II. End item visual defects. (cont'd)

	Examine	Defect	Classification Major	Minor A	Minor B
	Hardware (cont'd)	Corroded area			X
		Burr or sharp edge which may cause injury in handling or damage to fabric	X		
		Grommets improperly set; two or more teeth exposed (except where grommet is set as specified but is on an area of unequal thickness which prevents uniform clinching)			X
		Clinched loosely allowing grommet to rotate in hole		X	
		Grommet set with washer on inside of end section or intermediate section in lieu of outside			X
		Grommet set excessively tight, cutting fabric:			
		- on liner	X		
		- on cover		X	
*		Roll of a grommet contains splits			X
*	Becket line	Omitted	X		
		Stitching not passing through cord and roof portion	X		
		Stitching sewn completely on cord in lieu of off and on cord and roof			X
		Cord not continuous			X
		Any beckets loose (i.e., not caught in stitching on either side)		X	
		Not continuous or spliced within 6 inches of grommet			X
		Spliced cord less than 10 feet in length		X	
		Splice not centered between beckets			X
		Splice not made as indicated on Drawing 5-4-688		X	
	Suspension system (pipe eye ends, chain and "S" hooks)	Any component in liner suspension system not type specified or improperly assembled	X		
		Thread at ends of pipe damaged or otherwise deformed	X		
		Any pipe-end and end-eye assembly misfit, i.e., "S" hooks fail to engage end eyes when end section or intermediate section is fully extended	X		

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TABLE II. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Rope components	Any cut, any break		X	
	Any free end of polyester rope not finished as specified			X
Seams and stitching	Open seam on stitch type 301:			
	- for more than 1/4 inch but not more than 1 inch			X
	- for more than 1 inch		X	
	Open seam on stitch type 401:			
	- one row of stitching open for 1 inch or less			X
	- one row of stitching open for more than 1 inch:			
	- on liner sections		X	
	- on cover			X
	- both rows of stitching open (not necessarily at the same location):			
	- on liner sections	X		
	- on cover		X	
	Open seam on stitch type 304:			
	- for more than 1/4 inch but not more than 1 inch			X
	- for more than 1 inch	X		
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken and are not overstitched or when two or more skipped or runoff stitches occur and are not overstitched. On double stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.			
	Raw edge caught in stitching:			
	- more than 1 inch but not more than 3 inches			X
	- more than 3 inches		X	
	Thread breaks:			
	- overstitched less than 1 inch on stitch type 401			X
	- overstitched less than 1 inch on stitch type 301			X
	- overstitched less than 3 stitches on stitch type 304			X

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TABLE II. End item visual defects. (cont'd)

	Examine	Defect	Classification Major	Minor A	Minor B
	Seams and stitching (cont'd)	NOTE: Thread breaks not overstitched shall be classified as open seams.			
		Skipped stitches overstitched less than 1 inch on stitch type 401 or 301; less than 3 stitches on type 304 (when two or more skipped stitches occur)			X
		NOTE: Skipped stitches not overstitched shall be classified as open seams.			
		Needle chews resulting in cut, tear, or hole (see cut, tear or hole)			
		One or more required rows of stitching omitted (except on boxstitching)	X		
	Boxstitching	Incomplete:			
		- one row of stitching omitted		X	
		- two or more rows of stitching omitted	X		
		Any boxstitching less than specified length by more than 1/8 inch			X
*	Stitching ends	Overstitched less than 1/2 inch on stitch type 301 (except where ends are held down by other stitching or turned under in a hem, or where stitching is performed automatically)			X
		Secured with less than three tying, overlapping or backstitches when automatic stitching is performed		X	
		Secured with less than 1/4 inch of backstitching, overstitching, or by an increased number of stitches per inch at the ends of stitch type 304			X
	Seams	Seam pleated:			
		- on liner sections		X	
		- on cover			X
	Seam type	Wrong seam type	X		
	Stitch type	Wrong stitch type	X		

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TABLE II. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)				
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches		X	
	Tight, as evidenced by puckering on seams for more than 6 inches		X	
Stitches per inch	Less than minimum specified (see 3.3.1.1):			
	- one stitch			X
	- two or more stitches	X		
	More than maximum specified:			
	- one stitch			X
	- two or more stitches	X		
	NOTE: Variation in the number of stitches per inch, caused by the operators speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or in turning corners shall be classified as follows: a. Within the minor "B" defect classification - no defect b. Within the minor "A" defect classification - minor "B" defect			
Stitching gage	Not as specified			X
Stitching margin	Larger than specified:			
	- for more than 2 inches but not more than 4 inches in length			X
	- for more than 4 inches in length		X	
	Less than specified:			
	- for more than 2 inches but not more than 4 inches in length			X
	- for more than 4 inches in length		X	
Stitching indicated by "Z" on drawing	Chain portion of stitching visible outside of liner sections or cover	X		

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TABLE II. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)				
Mends, darns or patches	Not authorized by the contracting officer or not in accordance with general procedures as outlined in Department of the Army Field Manual FM 10-16 (see 3.3.8)	X		
Splices	More than one splice per panel. Location of splices, length of spliced pieces, or method of splicing not as specified	X		
Tufting	One or more tufts omitted	X		
	Any tufting done with less than two strands of 5-ply twine; ends of stitching not securely knotted		X	
	Tufting tape (or cloth) omitted on inside or outside face of sections		X	
Hems	Twisted or pleated			X
	Constructed with partial turn-under when a double turn-under is required		X	
Reinforcements	Improperly applied causing excessive fullness on reinforcement or reinforced part		X	
	Not stitched to reinforced part on all sides, or not securely caught in seams or stitching of reinforced part		X	
* Slide fasteners	Slider jams or fails to interlock chain	X		
	Any tape improperly assembled causing failure of chain to effect a smooth and secure closure	X		
	Wrong length	X		
	Width not as specified	X		
	Puckering of tape	X		
	Broken or missing scoops	X		
	Thong omitted	X		
	NOTE: The inspector shall pull slide fasteners three or four times to an open and closed position to determine proper function and effectiveness of each fastener.			

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TABLE II. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Components and assembly	Any component part not fabricated of applicable referenced material	X		
	Any operation omitted or improperly performed	X		
*	Synthetic cord, webbing or tape not fused as required		X	
Insulation	Not laid adjacent to seams		X	
	Buckled, not flat within shell, or doubled up, i.e., folded under			X
Labels	Incorrect, illegible, or stitched to outside of liner rather than inside		X	
	Identification, instruction, or warning label omitted		X	
Marking: "US" and identification	Omitted, incorrect, illegible, or misplaced			X
Cleanness	Stains (oil, grease, ink, dirt, etc.) clearly noticeable			X
	Thread end not trimmed to 1/4 inch or less			X

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TABLE III. End item visual defects (End section).

Examine	Defect	Classification Major	Minor A	Minor B
Openings	Ventilator opening and center of ventilator openings not reinforced with 1 inch tape as specified on drawings		X	
	Drum cover openings on both sides of ventilator not constructed as specified on Drawing 5-4-689, sections FF and GG		X	
	Drum cover openings not fully covered by drum cover openings as specified on drawings	X		
Ventilator flap	Fails to cover opening when in a closed position as specified on drawings	X		
	Stitched to top of openings instead of bottom			X
Right end wall Heater duct openings	Cut out in lower left end wall instead of lower right end wall; not securely reinforced all around	X		
Left end wall Personnel door opening	Cut out on right end wall instead of left end wall; not formed as specified	X		
* End section "A"	Becket loops omitted on free edge of roof section; becket loops extending from outer shell of roof section	X		
	Extending from outer shell (outside of blanket) in lieu of inner shell; omitted or furnished on both edges of blankets	X		
End section "B"	Roof section furnished with becket loops	X		

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TABLE IV. End item visual defects (Intermediate section).

Examine	Defect	Classification Major	Minor A	Minor B
Window opening	Not reinforced all around with 2-inch facing		X	
Window pane	Any puncture, deep crease, or crack	X		
	Pane not fully caught in binding stitching all around		X	
Ends of blanket	Ends of outer shell of blanket not flush with outer edge of hem on inner shell; ends of insulation caught in stitching forming hems		X	

* 4.4.4 End item dimensional examination.

* 4.4.4.1 End sections "A" or "B" and intermediate section. The end item, as applicable, shall be in a position so that both the inside and outside of the sections can be examined for defects listed in table V. The lot size shall be expressed in units of sections. The sample unit shall be one section. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15 for major and minor A combined defects and 40 for total (major and minor combined) defects.

* 4.4.4.2 Covers only. The end item, shall be in a position so that both the inside and outside of the cover can be examined for applicable defects listed in table V. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 for major defects, 15 for major and minor A combined defects and 65 for total (major and minor combined) defects.

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TABLE V. End item dimensional defects.

Examine	Defect	Classification Major	Minor A	Minor B
Overall dimensions of end sections "A" and "B" and intermediate sections	Smaller than specified dimensions less applicable minus tolerance indicated on drawings	X		
	Larger than specified dimensions and applicable plus tolerance indicated on drawings			X
End roof section	Ends of end roof section uneven with bottom edge of right and left end walls:			
	- by 1/2 inch or more, but not more than 1 inch		X	
	- by more than 1 inch	X		
Dimensions of cover	Smaller than specified dimensions less applicable minus tolerance indicated on drawings, but not smaller than specified dimensions less twice the applicable minus tolerance			X
	Smaller than specified dimensions less twice the applicable minus tolerance indicated on drawings	X		
	Larger than specified dimensions and applicable plus tolerance indicated on drawings			X
Vehicle door "A" and "B" openings	Length or width of openings more or less than specified by 1 inch or more		X	
	Out of specified location:			
	- by more than 1/2 inch but not more than 1 inch			X
	- by more than 1 inch	X		

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TABLE V. End item dimensional defects. (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Tie tapes	Finished length less than specified by more than 1/2 inch			X
	Tie tapes misplaced by more than 1/2 inch		X	
Rope components	Finished length less than specified by more than 2 inches			X
Hems	Hems required to finish less than 3/4 inch in width, less than specified; hems required to finish 3/4 inch or more in width, less than specified by more than 1/4 inch			X
Location dimensions	Components set beyond specified tolerances		X	
Right end wall heater duct openings	Out of specified location by more than 1 inch	X		
	Under or over specified dimensions by more than 1/2 inch		X	
Intermediate section window opening	Vertical center line out of alignment with center line of blanket:			
	- by more then 1/2 inch but not more than 1 inch			X
	- by more than 1 inch	X		
	Two inch facing uneven with edge of opening by more than 1/4 inch		X	

* 4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components.
Content	Number per container is more or less than required.

4.5 Methods of inspection.

4.5.1 Thickness test. The thickness of the insulation shall be determined by a suitable gage or measuring device. Three determinations shall be made per sample unit and the results shall be reported as the average of the three determinations to the nearest 1/16 inch.

4.5.2 Compressional recovery test.

4.5.2.1 Test apparatus. The test apparatus shall consist of the following:

a. A box of proper size to hold specimen; the sides of the box shall be so constructed that when a straight edge is rested across the top of any two sides, the straight edge is parallel to the bottom of the box.
b. Straight edge.
c. Depth gage.
d. Aluminum cover plate weighing 0.9 grams per square inch.
e. Weight equivalent to 1 pound per square inch of specimen area.

4.5.2.2 Specimen. The specimen shall be no smaller than 4 by 4 inches, and shall be cut no nearer than 12 inches from any edge of the glass fiber batting.

4.5.2.3 Method. Allow specimen to relax on table with no weight on surface for 24 hours before testing for compressional recovery. Place specimen in box with cover plate on top of the specimen. Place straight edge across the top of box to serve as reference stop for depth gage. Determine initial thickness of specimen (thickness of specimen and cover plate minus thickness of cover plate

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equals thickness of specimen). Remove depth gage and straight edge. Compress specimen under a weight equivalent to 1 pound per square inch evenly distributed over the whole surface of the specimen. Allow load to remain for one full minute. Remove weight from specimen. Permit specimen to relax for 15 seconds. Repeat this loading procedure two more times. Measure thickness of specimen within 1 minute after weight has been removed for final time.

4.5.2.4 Calculation.

$$\frac{\text{Thickness of specimen after compression}}{\text{Initial thickness of specimen}} \times 100 = \text{Percent compression recovery}$$

A minimum of five specimens shall be tested for each determination of compression recovery.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each tent liner end section and intermediate section shall be neatly folded to fit within the cover provided. Prior to folding, all slide fasteners shall be closed and the ventilator flaps secured by means of the tie tapes. Care should be exercised to avoid folding the windows. One folded tent liner section shall be placed within a cover. Each cover shall then be securely closed with all cover ends or flaps neatly folded within the package.

5.1.2 Commercial. The tent liner sections shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Tent liner sections of one type only, preserved as specified in 5.1, shall be packed in a close fitting cleated-plywood box conforming to overseas type, style A, type 3 load of PPP-B-601. Skids shall be fabricated in accordance with the gross weight requirements in the box specification. Each shipping container shall be closed and reinforced with flat strapping in accordance with the appendix of the box specification.

5.2.2 Level B packing. One tent liner section, preserved as specified in 5.1, shall be wrapped in cotton osnaburg cloth conforming to class 2 of CCC-C-429, or burlap cloth conforming to class 3 of CCC-C-467. The wrapping shall be securely handsewn with type I or II, 16 ply cotton twine conforming to T-T-871 or type I, size No. 2 jute twine conforming to T-T-911, with approximately one stitch to the inch and every third stitch knotted, or

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alternatively, the seams of the wrap may be machine-sewn. A minimum of 5 inches of wrapping material or tubing shall be gathered together on each of the four corners and securely sewn into ears for handles. The polyester thread used to manufacture the tent liner sections may be used for effecting the machine sewn closure of the wrapping or tubing.

5.2.3 Commercial packing. The tent liner sections, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.1 Additional marking. The following shall be stenciled or imprinted on each shipping container in characters not less than 3/4 inch high:

SEE ERECTION INSTRUCTIONS SEWN TO INSIDE OF TENT COVER

6. NOTES

6.1 Intended use. The tent liner sections are designed to be used with the Tent, Frame Type, Maintenance, Medium and are intended to provide added protection for maintenance crews in extremely cold climates.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
* b. Type and number of sections and covers required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.5).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 "S" hook source. American Chain & Cable Co., Inc., York, PA 17403, "S" hooks, size 6, 11 or 103 have been found to satisfactorily meet the requirements of 3.2.13.

6.4 Eye ends and rope clamp source. Astrup Manufacturing Co., Cleveland, OH 44113, eye ends, style 8, 3/8 inch and style 9, 1/2 inch; and brass rope clamp No. 877, have been found to satisfactorily meet the requirements of 3.2.15 and 3.2.16 respectively.

* 6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

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6.6 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG No. 2882, relative to camouflage requirements for tents, shelters and subsidiary components. When amendment, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.8 Subject term (key word) listing.

Liner
Frame-type
Maintenance tent
Section
Tent

* 6.9 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:

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Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8340-0510

Review activities:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-41812H	TENT LINER SECTIONS, FRAME-TYPE MAINTENANCE, MEDIUM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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